



Worksheet W7: Area Under the Curve

"The integral is the exact area under a curve."

Chapter 7 · 10 Problems · Name: _____ Date: _____

Area Under a Curve = The Integral = Total Accumulation

Problem 1

Question: If you draw a graph of speed over time, what does the **area under the curve** represent?

💡 *Hint: Think about the odometer.*

Problem 2

A car drives at a constant speed of 60 mph for 3 hours. What is the area under the speed curve?

💡 *Hint: Area of a rectangle = base $\times$ height.*

Problem 3

A car accelerates from 0 to 60 mph over 3 hours at a constant rate. What is the area under the speed curve?

💡 *Hint: Area of a triangle = (base $\times$ height) $\div$ 2.*

Problem 4

Find the area under $y = 2x$ from $x = 0$ to $x = 5$.

💡 Hint: The area is a triangle. Base = 5, height = $2 \times 5 = 10$.

Problem 5

Find the area under $y = 2x + 3$ from $x = 0$ to $x = 4$.

💡 Hint: Area of a trapezoid = $\frac{(h_1 + h_2)}{2} \times \text{base}$.

Problem 6

Use the left-hand sum with **4 rectangles** (width = 1) to estimate the area under $y = x^2$ from $x = 0$ to $x = 4$.

Interval	0 to 1	1 to 2	2 to 3	3 to 4
Left Edge	$x = 0$	$x = 1$	$x = 2$	$x = 3$
Height (x^2)	0	1	4	9
Area	0	1	4	9

💡 Hint: Add the areas: $0 + 1 + 4 + 9 = ?$

Problem 7

Use the right-hand sum with **4 rectangles** (width = 1) to estimate the area under $y = x^2$ from $x = 0$ to $x = 4$.

Interval	0 to 1	1 to 2	2 to 3	3 to 4
Right Edge	$x = 1$	$x = 2$	$x = 3$	$x = 4$
Height (x^2)	1	4	9	16
Area	1	4	9	16

💡 Hint: Add the areas: $1 + 4 + 9 + 16 = ?$

Problem 8

Question: What happens to the rectangle estimate as you use **more** and **thinner** rectangles?

💡 Hint: Does the estimate get closer or farther from the exact area?

Problem 9

Question: Complete this sentence: "The integral is the exact _____ under the curve."

💡 Hint: It's the space inside a shape.

Problem 10

If flow rate is 2 gallons per minute for 5 minutes, what is the **area under the flow rate curve**?

💡 Hint: Area = rate $\times$ time. The curve is a rectangle.